

Homework 6

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Problem 5.6

83/100

$$T = 300 \text{ K}$$

a. $D = 50 \text{ nm}$

$$A_c = 5 \text{ mm}^2$$

$$k = 2.3 \text{ W/m}\cdot\text{K}$$

$$h = 50 \text{ W/m}^2\cdot\text{K}$$

$$A_c = \pi r_0^2$$

$$r_0 = \sqrt{\frac{A_c}{\pi}}$$

$$V = A_c \pi D$$

$$A_x = 2\pi r_0 \pi D$$

$$A_x = 2\pi \sqrt{\frac{A_c}{\pi}} \pi D$$

$$L_c = \frac{V}{A_x} = \frac{A_c \pi D}{2\pi \sqrt{\frac{A_c}{\pi}} \pi D}$$

$$L_c = \frac{1}{2} \sqrt{\frac{A_c}{\pi}} = \frac{1}{2} \sqrt{\frac{5}{\pi}}$$

$$L_c = 0.631 \text{ mm}$$

$$L_c = r_0$$

$$= \sqrt{\frac{A}{\pi}} = \sqrt{\frac{5}{\pi}} = 1.26$$

$$Bi = \frac{h L_c}{k} = \frac{50 \times 0.631 \times 10^{-3}}{2.3}$$

$$Bi = 0.0137$$

$$0.0137 < 0.1$$

$\therefore$ The lumped capacitance approximation is valid.

b)

$$W = 3 \text{ mm}$$

$$H = 5 \text{ mm}$$

$$L = 100 \text{ mm}$$

$$h = 15 \text{ W/m}^2\cdot\text{K}$$

$$V = W H L$$

$$A_x = 2(W + H)L + 2WH$$

$$L_c = \frac{W H L}{2(W + H)L + 2WH}$$

$$L_c = \frac{5 \times 3 \times 100}{2(5 + 3)100 + 2 \times 5 \times 3} = 0.92$$

$$Bi = \frac{h L_c}{k} = \frac{15 \times 0.92 \times 10^{-3}}{14.9} = 0.00926$$

$Bi < 0.1$ lumped capacitance approximation is valid

(e) $D = 13 \text{ mm}$
 $K = 130 \text{ W/m}\cdot\text{K}$
 $T_{\text{wall}} = 18^\circ\text{C} = 291.15 \text{ K}$
 $T_i = 100^\circ\text{C} = 373 \text{ K}$
 $\varepsilon = 0.75$

$$V = \frac{\pi D^3}{6}$$

$$A_x = \pi D^2$$

$$L_c = \frac{\pi D^3/6}{\pi D^2}$$

$$L_c = \frac{D}{6} = \frac{13}{6} = 2.17 \text{ mm} \quad \checkmark$$

$$L'_c = \frac{D}{2} = \frac{13}{2} = 6.5 \text{ mm} \quad \checkmark$$

$$h_r = \varepsilon \sigma (T_i + T_{\text{wall}})(T_i^2 + T_{\text{wall}}^2)$$

$$h_r = 0.75 \times (5.67 \times 10^{-8}) (373 + 291)(373^2 + 291^2)$$

$$h_r = 6.41 \text{ W/m}^2\cdot\text{K}$$

$$Bi = \frac{6.41 \times (2.17 \times 10^{-3})}{130} = 0.000107 \quad \checkmark$$

$$Bi = \frac{6.4 \times 0.0064}{130} = 0.000315 \quad \checkmark$$

The value of $Bi < 0.1$ in both cases, the lumped capacitance approximation is valid

(f) $D = 20 \text{ mm}$
 $\rho = 2300 \text{ kg/m}^3$
 $C_p = 1750 \text{ J/kg}\cdot\text{K}$
 $K = 16 \text{ W/m}\cdot\text{K}$
 $T_\infty = 20^\circ\text{C}$
 $T_i = 200^\circ\text{C}$
 $T = 100^\circ\text{C}$ at $t = 225 \text{ s}$

$$V = \frac{\pi D^2 L}{4}$$

$$A_x = \pi D L$$

$$L_c = \frac{\pi D^2 L/4}{\pi D L} = \frac{D}{4} = \frac{20}{4} = 5$$

$$L'_c = \frac{D}{2} = \frac{20}{2} = 10$$

$$h = \frac{\rho D_c}{t A_x} \ln \frac{\theta_i}{\theta} = \frac{\rho D_c}{4t} \ln \left(\frac{T_i - T_\infty}{T - T_\infty} \right)$$

$$h = \frac{2300 \times 0.020 \times 1750}{4 \times 225} \ln \left(\frac{200 - 20}{100 - 20} \right)$$

$$h = 72.5$$

$$0.01 = 0.954 \times (\sqrt{\alpha \times 120})$$

$$\sqrt{\alpha \times 120} = \frac{0.01}{0.954}$$

$$\sqrt{\alpha \times 120} = 0.01048$$

$$\alpha = 9.15 \times 10^{-7} \text{ m}^2/\text{s}$$

$$\alpha = \frac{k}{\rho c}$$

$$k = \alpha \rho c$$

$$k = (9.15 \times 10^{-7} \text{ m}^2/\text{s}) (2200 \text{ kg/m}^3) (700 \text{ J/kg} \cdot \text{K})$$

$$k = 1.409 \text{ W/m} \cdot \text{K}$$

④ Problem 6.34

$$L = 800 \text{ mm}$$

$$t = 6 \text{ mm}$$

$$V = 70 \text{ mph}$$

$$T_{\infty,0} = -15^\circ\text{C}$$

$$\overline{Nu}_L = 0.030 Re_L^{0.8} Pr^{1/3}$$

$$Re_L = VL/\nu$$

$$k = 0.023 \text{ W/m} \cdot \text{K}$$

$$\nu = 12.5 \times 10^{-6} \text{ m}^2/\text{s}$$

$$Pr = 0.71$$

$$T_{dp} = 10^\circ\text{C}$$

$$T_{\infty,i} = 50^\circ\text{C}$$

$$q_i = q_o$$

$$\frac{T_{\infty,i} - T_{s,i}}{R_{t,con}} = \frac{T_{s,i} - T_{\infty,0}}{R_{tot}}$$

$$\frac{T_{\infty,i} - T_{s,i}}{1/\bar{h}_i A} = \frac{T_{s,i} - T_{\infty,0}}{R_{t,cond} + R_{t,conv}}$$

$$\frac{T_{\infty,t} - T_{s,t}}{1/\bar{h}_i A} = \frac{T_{s,t} - T_{\infty,0}}{t/k_g A + 1/\bar{h}_o A}$$

$$\frac{T_{\infty,i} - T_{s,i}}{1/\bar{h}_i} = \frac{T_{s,i} - T_{\infty,0}}{t/k_g + 1/\bar{h}_o}$$

$$\bar{h}_i = \left(\frac{T_{s,t} - T_{\infty,0}}{T_{\infty,t} - T_{s,t}} \right) \left[\frac{1}{t/k_g + 1/\bar{h}_o} \right]$$

$$V = 70 \times (1585) (1 \text{ hr} / 3600 \text{ s})$$

$$V = 30.82 \text{ m/s}$$

$$Re = \frac{VL}{\nu} = \frac{30.82 \times 800 \times 10^{-3}}{12.5 \times 10^{-6}} = 1972480$$

$$\overline{Nu}_L = 0.03 Re_L^{0.8} Pr^{1/3}$$

$$\frac{\bar{h}_o L}{k} = 0.03 Re_L^{0.8} Pr^{1/3}$$

$$\frac{\bar{h}_o \times 800 \times 10^{-3}}{0.023} = 0.03 \times (1972480)^{0.8} (0.71)^{1/3}$$

$$\bar{h}_o = 83.59 \text{ W/m}^2 \cdot \text{K}$$